

Numerical Analysis – Homework #1 (Solutions)

Utah State University Fall 2026

1.

Consider the function $f(x) = \sin x$ near $x = 0$.

(a) Write the Taylor polynomials $P_1(x)$, $P_3(x)$, and $P_5(x)$.

Solution: About $x = 0$:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

Hence

$$P_1(x) = x, \quad P_3(x) = x - \frac{x^3}{6}, \quad P_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}.$$

(b) Graph $P_1(x)$, $P_3(x)$, and $P_5(x)$ as well as $f(x)$ over $[0, 2\pi]$.

Solution: The plots show P_5 closely tracking $\sin x$ on $[0, 2\pi]$, with accuracy degrading near the interval ends; P_3 improves on P_1 , and P_1 is tangent at 0 but quickly diverges for larger x .

(c) Compute $P_1(0.1)$, $P_3(0.1)$, and $P_5(0.1)$ and find the absolute and relative error.

Solution: $f(0.1) = \sin(0.1) \approx 0.09983341664682815$.

$$P_1(0.1) = 0.1, \quad |E| = 1.6658 \times 10^{-4}, \quad \text{rel} = 1.6686 \times 10^{-3},$$

$$P_3(0.1) = 0.09983333333333334, \quad |E| = 8.3313 \times 10^{-8}, \quad \text{rel} = 8.3453 \times 10^{-7},$$

$$P_5(0.1) = 0.09983341666666667, \quad |E| = 1.9839 \times 10^{-11}, \quad \text{rel} = 1.9872 \times 10^{-10}.$$

(d) Compute $P_1(0.5)$, $P_3(0.5)$, and $P_5(0.5)$ and find the absolute and relative error.

Solution: $f(0.5) = \sin(0.5) \approx 0.479425538604203$.

$$P_1(0.5) = 0.5, \quad |E| = 2.0574 \times 10^{-2}, \quad \text{rel} = 4.2915 \times 10^{-2},$$

$$P_3(0.5) = 0.4791666666666667, \quad |E| = 2.5887 \times 10^{-4}, \quad \text{rel} = 5.3996 \times 10^{-4},$$

$$P_5(0.5) = 0.47942708333333334, \quad |E| = 1.5447 \times 10^{-6}, \quad \text{rel} = 3.2220 \times 10^{-6}.$$

(e) Compute $P_1(1)$, $P_3(1)$, and $P_5(1)$ and find the absolute and relative error.

Solution: $f(1) = \sin(1) \approx 0.8414709848078965$.

$$P_1(1) = 1, \quad |E| = 1.58529 \times 10^{-1}, \quad \text{rel} = 1.88395 \times 10^{-1},$$

$$P_3(1) = 0.8333333333333334, \quad |E| = 8.13765 \times 10^{-3}, \quad \text{rel} = 9.6707 \times 10^{-3},$$

$$P_5(1) = 0.8416666666666667, \quad |E| = 1.95682 \times 10^{-4}, \quad \text{rel} = 2.3255 \times 10^{-4}.$$

2.

Consider the function $f(x) = \ln(1+x)$ near $x = 0$.

(a) Write the Taylor polynomials $P_1(x)$, $P_2(x)$, and $P_3(x)$.

Solution:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

$$\text{Hence } P_1(x) = x, \quad P_2(x) = x - \frac{x^2}{2}, \quad P_3(x) = x - \frac{x^2}{2} + \frac{x^3}{3}.$$

(b) Graph $P_1(x)$, $P_2(x)$, $P_3(x)$ and $f(x)$ over $[0, 2]$.

Solution: Over $[0, 2]$, P_3 is best; P_1 overestimates $\ln(1+x)$ while P_2 underestimates for larger x . Accuracy is excellent near 0 and degrades as x grows.

(c) Compute $P_1(0.1)$, $P_2(0.1)$, and $P_3(0.1)$ and find the absolute and relative error.

Solution: $f(0.1) = \ln(1.1) \approx 0.09531017980432493$.

$$P_1(0.1) = 0.1, \quad |E| = 4.6898 \times 10^{-3}, \quad \text{rel} = 4.9206 \times 10^{-2},$$

$$P_2(0.1) = 0.095, \quad |E| = 3.1018 \times 10^{-4}, \quad \text{rel} = 3.2544 \times 10^{-3},$$

$$P_3(0.1) = 0.09533333333333334, \quad |E| = 2.3154 \times 10^{-5}, \quad \text{rel} = 2.4293 \times 10^{-4}.$$

(d) Compute $P_1(0.5)$, $P_2(0.5)$, and $P_3(0.5)$ and find the absolute and relative error.

Solution: $f(0.5) = \ln(1.5) \approx 0.4054651081081644$.

$$P_1(0.5) = 0.5, \quad |E| = 9.4535 \times 10^{-2}, \quad \text{rel} = 2.3315 \times 10^{-1},$$

$$P_2(0.5) = 0.375, \quad |E| = 3.0465 \times 10^{-2}, \quad \text{rel} = 7.5136 \times 10^{-2},$$

$$P_3(0.5) = 0.4166666666666667, \quad |E| = 1.1202 \times 10^{-2}, \quad \text{rel} = 2.7626 \times 10^{-2}.$$

(e) Compute $P_1(1)$, $P_2(1)$, and $P_3(1)$ and find the absolute and relative error.

Solution: $f(1) = \ln 2 \approx 0.6931471805599453$.

$$\begin{aligned} P_1(1) &= 1, & |E| &= 3.0685 \times 10^{-1}, & \text{rel} &= 4.42695 \times 10^{-1}, \\ P_2(1) &= 0.5, & |E| &= 1.9315 \times 10^{-1}, & \text{rel} &= 2.7865 \times 10^{-1}, \\ P_3(1) &= 0.8333333333333333, & |E| &= 1.4019 \times 10^{-1}, & \text{rel} &= 2.0225 \times 10^{-1}. \end{aligned}$$

3.

Construct a Taylor polynomial approximation to $f(x) = e^x$ with $x_0 = 0$ that is accurate to within 10^{-3} on the interval $[0, 1]$. Find a value M such that

$$|f(x_1) - f(x_2)| \leq M|x_1 - x_2|$$

for all x_1, x_2 on the interval. (Use the Mean Value Theorem.)

Solution: The Maclaurin polynomial of degree n has remainder

$$R_{n+1}(x) = \frac{e^\xi}{(n+1)!} x^{n+1}, \quad \xi \in (0, x).$$

On $[0, 1]$, $e^\xi \leq e$ and $x^{n+1} \leq 1$, so $|R_{n+1}(x)| \leq \frac{e}{(n+1)!}$. The smallest n with $\frac{e}{(n+1)!} \leq 10^{-3}$ is $n = 6$ (since $e/7! \approx 5.39 \times 10^{-4}$). Thus the degree-6 Maclaurin polynomial achieves the accuracy.

By the Mean Value Theorem, for some ξ between x_1 and x_2 ,

$$|e^{x_1} - e^{x_2}| = e^\xi |x_1 - x_2| \leq e |x_1 - x_2| \quad (x \in [0, 1]).$$

Hence one can take $M = e$.

4.

Find the Taylor polynomials of order 2 and order 3 around $x_0 = 1$ for

$$f(x) = x^3 + x.$$

Solution: Compute derivatives: $f'(x) = 3x^2 + 1$, $f''(x) = 6x$, $f^{(3)}(x) = 6$. At $x_0 = 1$:

$$f(1) = 2, \quad f'(1) = 4, \quad f''(1) = 6, \quad f^{(3)}(1) = 6.$$

Let $h = x - 1$. Then

$$P_2(x) = 2 + 4h + \frac{6}{2}h^2 = 2 + 4(x - 1) + 3(x - 1)^2,$$

$$P_3(x) = P_2(x) + \frac{6}{6}h^3 = 2 + 4(x - 1) + 3(x - 1)^2 + (x - 1)^3.$$

5.

Find intervals containing solutions to the following equations:

(a) $x - 3^{-x} = 0$

Solution: Let $g(x) = x - 3^{-x}$. Then $g(0) = -1 < 0$ and $g(1) = 1 - \frac{1}{3} > 0$, so there is a root in $(0, 1)$.

(b) $4x^2 - e^x = 0$

Solution: $h(0) = -1 < 0$ and $h(1) = 4 - e > 0$, so one root lies in $(0, 1)$. For large x , e^x dominates; check $h(4) = 64 - e^4 > 0$ and $h(5) = 100 - e^5 < 0$, hence a second root in $(4, 5)$.